

Name

Roll No.:

General Instructions:

1. Read the questions carefully.
2. Attempt all the questions.
3. No negative marking.
4. Answers to be marked in the OMR sheet. Darken the bubble completely. Partially darkened bubbles will not be considered.
5. Question number 1 to 100 carries 1 mark each

Read the given passage and answer the questions (1-4) that follow:

Necessity, they say, is the mother of invention. And Indian doctors have been quite creative when hamstrung by few or no tools to perform specific surgeries. Their ability to innovate under constraints has often led to devices that are both cost-effective and highly efficient. They simply design it themselves at one-fourth the price they are sold abroad. In fact, some of their innovations are priced at as much as hundreds of dollars abroad.

Take 47-year-old Dr. Burjor P. Banaji, pioneer of Lasik surgery in India. He's invented over a dozen surgical instruments. When this senior eye surgeon at Max Eye Care started Lasik, there were few surgeons doing it worldwide and no specific instruments were available either.

"As I want things super-perfect, I designed a whole slew of instruments that made my surgery more efficient," says Banaji. The most popular instruments are Banaji Lasik Shield and Banaji Lasik Spatula and Canulae.

"It was simple. I had the designs in my head. Putting them down on paper was the simplest thing," he says. Instruments manufacturers and large multinationals in the US snapped them up. "They would send me computer generated drawings which I would correct and send back. Their level of execution was astounding. Within two weeks of the designs being finalised, the instruments were in the world market."

His instruments are priced at hundreds of dollars each in the US, and are also sold in Switzerland, South America, Korea, Eastern Europe, Africa and Japan. This wide acceptance highlights how Indian medical innovations can match global standards. They're available in India at a fraction of the price.

1. What does the passage suggest about Indian doctors when they face a lack of surgical tools?
 - a) They wait for instruments to arrive from abroad.
 - b) They stop performing the surgery.
 - c) They invent and design their own instruments.
 - d) They refuse to lower the cost of instruments.
2. Which of the following statements is **NOT** true according to the passage?
 - a) Dr. Banaji's inventions include the Lasik Shield, Spatula, and Canulae.
 - b) His surgical instruments are priced at hundreds of dollars in the US and are also sold in Japan, Korea, and Africa.
 - c) The instruments are sold in India at a fraction of the international price.
 - d) The instruments are costlier in India than in Switzerland or South America.
3. The passage explains that Dr. Banaji's designs entered the global market swiftly because manufacturers in the US provided him with _____ which he _____ before sending back.
 - a) surgical instruments ... modified physically
 - b) computer-generated drawings ... refined conceptually
 - c) technical blueprints ... copied carefully
 - d) medical samples ... tested experimentally
4. What does the word "**slew**" mean in the sentence: "**As I want things super-perfect, I designed a whole slew of instruments that made my surgery more efficient**"?
 - a) A variety
 - b) A large number
 - c) An assortment
 - d) A collection

5. Read the sentences given below:

1. She enjoys reading storybooks.
2. He decided to read the novel again.
3. The reading lamp was glowing softly.

Which option correctly identifies the underlined **verbal form** in each sentence?

- | | |
|---|---|
| a) (1) Gerund, (2) Infinitive, (3) Participle | b) (1) Infinitive, (2) Gerund, (3) Participle |
| c) (1) Participle, (2) Infinitive, (3) Gerund | d) (1) Gerund, (2) Participle, (3) Infinitive |

6. Which of the following sentences shows an incorrect use of the **comparative degree**?

- a) This book is more interesting than the one I read yesterday.
- b) Ramesh is more taller than his brother.
- c) She is more careful with her work than before.
- d) Of the two bags, the blue one is more expensive.

7. Choose the option that is the **closest antonym of fraught**.

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| a) Empty | b) Calm | c) Relaxed | d) Heavy |
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8. In which of the following options is the **subordinate clause** accurately identified?

- a) I know that she is honest. – Adjective Clause
- b) The place where he was born is famous. – Noun Clause
- c) He left because he was tired. – Adverb Clause
- d) The book that you gave me is interesting. – Noun Clause

9. What does the word **capital** mean in the sentence: '**The company raised additional capital to expand its operations.**'

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|---|---|
| a) Staff or workers employed in a company | b) A city chosen to control operations easily |
| c) Money or assets used for investment | d) Wealth in the form of ideas or knowledge that adds value |

10. In the sentence "**She was anxious to avoid the noisy crowd, so she eluded the market by taking a longer route,**" which word has been used inappropriately?

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|------------|----------|-----------|-----------|
| a) Anxious | b) Avoid | c) Eluded | d) Market |
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11. In which of the following words is "**re-**" NOT actually used as a prefix meaning "**again/back**"?

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|------------|---------------|------------|------------|
| a) Retreat | b) Reconsider | c) Rebound | d) Replace |
|------------|---------------|------------|------------|

12. In which of the following sentences is the underlined verb accurately identified?

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|--|--|
| a) She <u>can</u> swim across the river. – Main verb | b) They have <u>completed</u> their homework. – Auxiliary verb |
| c) He <u>must</u> obey the traffic rules. – Modal verb | d) She is a <u>doctor</u> . – Auxiliary verb |

13. Which of the following transformations is incorrect?

- a) Assertive: He is the best player in the team.
Interrogative: Isn't he the best player in the team?
- b) Positive: The task is possible to complete.
Negative: The task is not impossible to complete.
- c) Assertive: Everyone likes her singing.
Interrogative: Who does not like her singing?
- d) Positive: She is always punctual.
Negative: She is never punctual.

14. Identify the option in which the **contraction** is expanded correctly.

- a) She'd finished her homework before dinner. → She had finished her homework before dinner.
- b) He'd enjoy the play if he had tickets. → He had enjoy the play if he had tickets.
- c) They're planning a trip this weekend. → They were planning a trip this weekend.
- d) I'd rather stay home today. → I had rather stay home today.

15. Identify the sentence in which the **modal verb** is used incorrectly.

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| a) You must submit the form before the deadline. | b) He can play the piano since he was five years old. |
| c) We should respect our elders. | d) They might arrive later if the train is delayed. |

16. Identify the sentence in which the underlined word is used in its **sports-related sense**.
- The coach asked him to pitch his idea at the meeting.
 - She trained hard to improve her stroke in swimming.
 - The company is planning to sprint ahead of its competitors this year.
 - The lawyer made a strong defense in court.
17. Identify the sentence in which the **adverb** is used correctly:
- She sings more sweet than her sister.
 - He almost ran five miles without stopping.
 - The teacher spoke very loudly enough for everyone to hear.
 - They completed the project most quickly than all the teams.
18. Match the sentences with the correct tense type (one mismatch is hidden – find the correct sequence):
- She will have been working here for five years by December.*
 - They had been waiting for hours before the gates opened.*
 - He writes in his diary every evening.*
 - I was reading when the phone rang.*
- Future Perfect Continuous – Past Perfect Continuous – Present Simple – Past Continuous
 - Future Perfect – Past Perfect – Present Continuous – Past Simple
 - Future Perfect Continuous – Past Simple – Present Perfect – Past Continuous
 - Future Continuous – Past Perfect Continuous – Present Simple – Past Perfect
19. Read the passage below. Which option shows the correct punctuation?
- Riya asked have you read the article titled the future of AI it was published in yesterdays newspaper I think its very interesting dont you**
- Riya asked, "Have you read the article titled The Future of AI? It was published in yesterday's newspaper, I think it's very interesting don't you."
 - Riya asked, "Have you read the article titled The Future of AI? It was published in yesterday's newspaper. I think it's very interesting, don't you?"
 - Riya asked, "Have you read the article titled the future of AI? It was published in yesterday's newspaper? I think its very interesting don't you."
 - Riya asked "Have you read the article titled The Future of AI it was published in yesterdays newspaper I think it's very interesting don't you."
20. Choose the option that correctly completes the sentence using a suitable **conjunction**:
- "He decided to join the team, _____ he knew the project would be very challenging."
- although
 - because
 - therefore
 - lest
21. Bobby travels 6metres North and then 8 metres East. What is his distance from starting point ?
- 10 metres
 - 14 metres
 - 2 metres
 - 7 metres
22. $(1 - \frac{1}{2})(1 - \frac{1}{3})(1 - \frac{1}{4})(1 - \frac{1}{5})(1 - \frac{1}{6})(1 - \frac{1}{7})(1 - \frac{1}{8})(1 - \frac{1}{9})(1 - \frac{1}{10}) = \underline{\hspace{2cm}}$
- $\frac{9}{10}$
 - 9^2
 - $\frac{1}{10}$
 - $\frac{1}{2}$
23. A pen and a pencil together cost Rs.20. The pen costs Re.1 more than the cost of pencil. Find the cost of a pen.
- Rs. 9.50
 - Rs. 9
 - Rs.10
 - Rs. 10.50
24. A is two years older than B, who is twice as old as C. If total ages of A, B and C be 27, how old is B?
- 5
 - 8
 - 10
 - 12
25. 75% of a number is added to 75, then the result is the number itself. What is the number?
- 50
 - 60
 - 300
 - 400
26. The prices of a scooter and TV are in the ratio 7 : 5. If scooter costs Rs. 8,000 more than TV set, then price of TV set is :
- Rs. 20,000
 - Rs. 24,000
 - Rs. 28,000
 - Rs. 32,000

27. What is 90% of 90% of 100 ?
 a) 63 b) 72 c) 81 d) 90
28. $\frac{5^n 25^{n-1}}{5^{n-1} 25^{n-1}} = \text{-----}$.
 a) 0 b) 20 c) 25 d) 5
29. Which of the following numbers is divisible by both 3 and 11?
 a) 495 b) 308 c) 874 d) 704
30. The product of two consecutive even numbers is 288. What are the numbers?
 a) 14, 16 b) 16, 18 c) 18, 20 d) 12, 14
31. The area of a triangle is 30 cm² and its base is 10 cm. What is the height?
 a) 3 cm b) 5 cm c) 6 cm d) 8 cm
32. If $5x - 7 = 18$, then $3x + 2 = ?$
 a) 12 b) 15 c) 17 d) 20
33. The sum of the angles of a triangle is 180°. If one angle is 3 times another and the third angle is 40°, what is the smallest angle?
 a) 25° b) 30° c) 35° d) 40°
34. A number when divided by 5 leaves a remainder 3. What will be the remainder when the square of this number is divided by 5?
 a) 1 b) 2 c) 3 d) 4
35. The perimeter of a regular hexagon is 72 cm. Find the length of each side.
 a) 10 cm b) 11 cm c) 12 cm d) 15 cm
36. If $a = 3$ and $b = 4$, what is the value of $(a^2 + b^2) / (a + b)$?
 a) 7/5 b) 5/7 c) 25/7 d) 7/25
37. In a bar graph showing the number of books read by students in a month, the heights of bars for A, B, C, D are 5, 8, 6, and 9 cm respectively. Who read the second highest number of books?
 a) A b) B c) C d) D
38. The mode of numbers 4, 6, 7, 8, 9, 6, 8, 7, 8 is:
 a) 6 b) 7 c) 8 d) 9
39. A pie chart shows that 25% of students like Maths, 35% like Science, 20% like English, and 20% like Social Studies. If there are 200 students, how many like Science?
 a) 35 b) 60 c) 70 d) 80
40. The mean of 6 numbers is 12. If one number is removed and the new mean becomes 11, find the number removed.
 a) 18 b) 17 c) 14 d) 6
41. Which natural disaster is characterized by a rotating column of air extending from a thunderstorm to the ground?
 a) Hurricane b) Tornado c) Earthquake d) Tsunami
42. Which natural disaster is often triggered by heavy rainfall or snowmelt?
 a) Earthquake b) Landslide c) Hurricane d) Tsunami
43. What is the primary purpose of sewage treatment plants?
 a) To remove solid waste b) To purify water for drinking
 c) To reduce water pollution d) To increase water supply
44. Which method is commonly used to remove suspended solids from wastewater?
 a) Sedimentation b) Filtration c) Chlorination d) Aeration

45. What is the primary function of the reproductive system in living organisms?
a) To produce energy b) To support growth and development
c) To ensure continuity of species d) To regulate body functions
46. Which type of reproduction involves the fusion of gametes?
a) Asexual reproduction b) Sexual reproduction c) Vegetative propagation d) Regeneration
47. What is the most important basis for classifying substances into acids and bases?
a) pH value b) Chemical composition c) Physical state d) Solubility in water
48. Which substance is a strong acid?
a) Acetic acid b) Hydrochloric acid c) Citric acid d) Tartaric acid
49. What is the phenomenon that occurs when white light passes through a prism?
a) Reflection b) Refraction c) Dispersion d) Diffraction
50. Which type of mirror is used in a solar furnace?
a) Plane mirror b) Concave mirror c) Convex mirror d) Spherical mirror
51. What percentage of the Earth's surface is covered with water?
a) 50% b) 70% c) 90% d) 100%
52. Which process is primarily responsible for the purification of water in nature?
a) Evaporation b) Condensation c) Transpiration d) Water cycle
53. What is the basic structural and functional unit of living organisms?
a) Cell b) Tissue c) Organ d) System
54. Which level of organization is characterized by a group of cells performing a specific function?
a) Cellular level b) Tissue level c) Organ level d) System level
55. What is the primary function of the circulatory system?
a) To transport oxygen and nutrients b) To regulate body temperature
c) To support immune function d) To remove waste products
56. Which blood vessel carries oxygenated blood from the lungs to the heart?
a) Pulmonary artery b) Pulmonary vein c) Aorta d) Superior vena cava
57. What is the pH value of a neutral solution?
a) 0 b) 7 c) 14 d) -1
58. Which salt is commonly used as a fertilizer?
a) Sodium chloride b) Ammonium sulfate c) Copper sulfate d) Potassium nitrate
59. What is the unit of measurement for specific heat capacity?
a) J/g°C b) J/kg°C c) cal/g°C d) kcal/kg°C
60. Which process involves the transfer of heat energy through direct contact?
a) Conduction b) Convection c) Radiation d) Evaporation
61. Which of the following best explains the economic impact of British rule on India?
a) Industrial growth increased
b) Traditional industries declined due to imports
c) Indian agriculture became self-sufficient
d) Indian rulers gained more power
62. Which of the following was the main reason for the conflict between the Vijayanagar and Bahmani Kingdoms?
a) Trade routes and fertile lands b) Religion c) Cultural differences d) foreign invasion

63. Which layer of the Earth is made up mostly of solid iron and nickel?
 a) Crust b) Mantle c) Outer core d) Inner core
64. A farmer uses land, labour, and capital to grow crops. What is he contributing to?
 a) The Constitution b) The factors of production
 c) The natural vegetation d) The defence system
65. The Nayakas were —
 a) Soldiers under the British Empire
 b) Local chieftains who ruled small territories
 c) Farmers under Mughal rule
 d) Traders from Europe
66. Akbar's policy of Sulh-i-Kul promoted —
 a) religious tolerance among communities b) Wars among kingdoms
 c) Increase in taxes d) Slavery system
67. Farmers in black soil regions mainly grow —
 a) Rice b) Cotton c) Tea d) Wheat
68. Why do citizens pay taxes?
 a) To punish the poor b) To support public services like roads and schools
 c) To reduce democracy d) To help only private companies
69. A city experiences hot summers and mild winters because it is near the sea. This is due to —
 a) Latitude b) Ocean currents c) Altitude d) Distance from the equator
70. Which policy of the British led to the loss of power of Indian kings?
 a) Divide and Rule b) Doctrine of Lapse c) Quit India d) Swadeshi
71. People build houses on stilts in coastal and flood-prone areas because —
 a) It is fashionable b) It looks attractive
 c) It prevents water from entering homes d) It is cheaper
72. Why was the Indian Constitution written after independence?
 a) To create rules for a free nation b) To celebrate victory
 c) To copy British laws d) To increase taxes
73. What was the main difference between nomads and settled communities?
 a) Nomads moved from place to place b) Nomads lived in big cities
 c) Settled people did not farm d) Both depended on hunting only
74. Why do farmers depend on monsoon rains in India?
 a) To build roads b) To grow crops and fill reservoirs
 c) To run industries d) To increase imports
75. Which mountain range runs along the western side of South America?
 a) Alps b) Andes c) Rockies d) Himalayas
76. The Bhakti movement emphasized—
 a) Idol worship only b) Ritual sacrifices
 c) Personal devotion to God d) Caste system
77. A student analysing monsoon patterns to plan crop cycles is applying knowledge of—
 a) Weather and Climate b) Political Science
 c) Industrial Development d) History
78. Which principle best supports democracy in a country?
 a) Censorship of media b) Concentration of power
 c) Fundamental Rights and Duties d) Rule by a single leader

79. Imagine you are part of the freedom struggle. What new method could you suggest to mobilize people peacefully?
- Violent protests
 - Passive resistance using public speeches and boycotts
 - Forming secret armies
 - Avoiding political discussions
80. If a new mineral resource is discovered in a region, which human activity is most likely to increase first?
- Tourism
 - Industrialization
 - Agriculture
 - Education
81. If $5 + 6 = 31$, $6 + 7 = 43$, $7 + 8 = 57$ then $8 + 9 = \underline{\quad\quad}$
- 69
 - 73
 - 80
 - 65
82. Find the odd one out of the sequence of numbers: 25, 27, 30, 35, 42, 53, 65
- 30
 - 42
 - 53
 - 65
83. A 1 km long train passes through a tunnel of 1 km length at a speed of 1 km per minute. How much time will it take to pass through it completely?
- 1 min
 - 2 min
 - 5 min
 - 0.5 min
84. If the day before yesterday was Saturday, what day will fall on the day after tomorrow?
- Friday
 - Thursday
 - Wednesday
 - Tuesday
85. Select a figure from amongst the Answer Figures which will continue the same series as established by the five Problem Figures.
- Problem Figures: Answer Figures:
- | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| □ | △ | ? | □ | = | ? | x | = | △ | x |
| ? | = | △ | x | □ | △ | △ | □ | □ | ? |
| x | ? | △ | x | □ | △ | ? | □ | = | ? |
- | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| △ | □ | □ | △ | = | △ | x | □ | △ | = |
| ? | = | x | ? | ? | x | ? | △ | x | ? |
| x | = | x | = | □ | x | ? | △ | x | ? |
- (A) (B) (C) (D) (E) (1) (2) (3) (4) (5)
- (1)
 - (2)
 - (3)
 - (4)
86. Pointing to a man, Deepak said, "His only brother is the father of my daughter's father." How is the man related to Deepak?
- Grandfather
 - Father
 - Uncle
 - Brother-in-law
87. Aruna ranks 12th in a class of 46. What will be her rank from the last?
- 33
 - 34
 - 35
 - 37
88. Complete the series: A, B, D, G, _____
- M
 - L
 - K
 - H
89. ACEG : DFHJ :: QSUW : ?
- KMNP
 - MNPR
 - TQST
 - TVXZ
90. In a certain code language 'POND' is written as 'RSTL'. How will 'HEAR' be written in that code?
- GHIZ
 - GHIJ
 - JCLZ
 - JIGZ
91. Who is the main character in The Hobbit?
- Frodo Baggins
 - Harry Potter
 - Bilbo Baggins
 - Gandalf
92. Which mineral makes our teeth and bones strong?
- Iron
 - Calcium
 - Potassium
 - Sodium
93. Which insect is known as a social insect?
- Beetle
 - Butterfly
 - Ant
 - Grasshopper
94. Which famous English poet wrote The Rime of the Ancient Mariner?
- William Wordsworth
 - Samuel Taylor Coleridge
 - John Keats
 - Percy Bysshe Shelley

95. Which Indian state is known as the "Granary of India"?
a) Punjab b) Gujarat c) Kerala d) Maharashtra
96. The movie Baahubali was originally made in which language?
a) Tamil b) Hindi c) Telugu d) Kannada
97. Which new sport debuted at the Paris 2024 Olympics
a) Skateboarding b) Sportclimbing
c) Breaking d) Karatte
98. The **Interim Budget 2025** of India was presented by —
a) Nirmala Sitharaman b) Piyush Goyal
c) Arun Jaitley d) Rajnath Singh
99. Which country is known as the "**Land of the Rising Sun**"?
a) China b) Japan c) Thailand d) South Korea
100. ISRO's **Aditya-L1 Mission** is designed to study —
a) The Moon b) Mars c) The Sun d) Jupiter